

AI FAQ ASSISTANT API

- **Project Title:** "AI FAQ Assistant API"
- **Team ID:** NM2025TMID38548
- **Team Leader:** M. Divya sree
- **Mail id:** sreed1347@gmail.com [Coder]
- **Team Members:**
 - NAME: M.Anusree
MAILID: anusreemuruganandham2007@gmail.com [Coder]
 - NAME: N.Deepika
MAIL ID: deepikapooja2728@gmail.com [Editing]
 - NAME: M.Dhivya sree
MAIL ID: dhivyaaleela30@gmail.com [Documenter]
 - NAME: A.Ambeeleswari
MAIL ID: ambeeleswari14@gmail.com [Documenter]

Project Testing Phase

AI FAQ Assistant API

The testing phase of the **AI FAQ Assistant API** is performed to ensure that the system works correctly, provides relevant answers, handles errors properly, and remains reliable under different conditions. Testing is carried out on the AI model, API endpoints, database, security, and overall system performance.

1. Functional Testing

Functional testing verifies whether the API performs its required functions correctly. Different FAQ questions are submitted, and the generated responses are compared with the expected answers. All major API endpoints are tested to ensure proper functionality.

2. AI Response Testing

The AI assistant is tested with questions having different wording but the same meaning. It is also tested with incomplete, ambiguous, and unrelated questions. This ensures that the system can understand user intent and provide suitable answers or an appropriate fallback response.

3. API Testing

API testing checks request and response formats, HTTP status codes, input validation, and error handling. Both valid and invalid requests are tested to confirm that the API responds correctly and consistently.

4. Database Testing

Database testing verifies that FAQ information is correctly stored, retrieved, updated, and deleted. It also ensures that changes made to the FAQ knowledge base are properly reflected in the assistant's responses.

5. Performance Testing

Performance testing measures response time and system behavior when multiple requests are received simultaneously. The objective is to ensure that the API can handle the expected number of users without significant delays.

6. Security Testing

Security testing is performed to protect the API and its data. Authentication, authorization, input validation, API keys, and rate limiting are tested. Unauthorized access and invalid requests are also checked.

7. Integration Testing

Integration testing verifies the communication between the API, AI model, database, and other required services. The complete process, from receiving a user question to returning the final answer, is tested as a single workflow.

8. User Acceptance Testing

The system is tested using realistic questions from potential users. Feedback is collected regarding response accuracy, relevance, speed, and ease of use. Necessary corrections are made based on the test results.

Conclusion

The testing phase confirms that the **AI FAQ Assistant API** can process user queries, retrieve relevant information, and provide appropriate responses effectively. Functional, AI response, API, database, performance, security, integration, and user acceptance testing help identify and correct potential problems.

